In[ ]:= **Quit[]**
detén núcleo del sistema

In[ ]:= **<< xAct`xTensor`**

```
------------------------------------------------------------
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Connecting to external MinGW executable...
Connection established.
------------------------------------------------------------
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------------------------------------------------------------
```

In[ ]:= **$PrePrint = ScreenDollarIndices;**
pre-escribe

In[ ]:= **DefManifold[M, 4, {a, b, c, d, e, f, m}]**

```
** DefManifold: Defining manifold M.
** DefVBundle: Defining vbundle TangentM.
```

In[ ]:= **DefMetric[-1, g[-a, -b], CD, PrintAs → "g"]**

```
** DefTensor: Defining symmetric metric tensor g[-a, -b].
** DefTensor: Defining antisymmetric tensor epsilong[-a, -b, -c, -d].
** DefTensor: Defining tetrametric Tetrag[-a, -b, -c, -d].
** DefTensor: Defining tetrametric Tetrag†[-a, -b, -c, -d].
** DefCovD: Defining covariant derivative CD[-a].
** DefTensor: Defining vanishing torsion tensor TorsionCD[a, -b, -c].
** DefTensor: Defining symmetric Christoffel tensor ChristoffelCD[a, -b, -c].
** DefTensor: Defining Riemann tensor RiemannCD[-a, -b, -c, -d].
** DefTensor: Defining symmetric Ricci tensor RicciCD[-a, -b].
** DefCovD: Contractions of Riemann automatically replaced by Ricci.
** DefTensor: Defining Ricci scalar RicciScalarCD[].
** DefCovD: Contractions of Ricci automatically replaced by RicciScalar.
** DefTensor: Defining symmetric Einstein tensor EinsteinCD[-a, -b].
** DefTensor: Defining Weyl tensor WeylCD[-a, -b, -c, -d].
** DefTensor: Defining symmetric TFRicci tensor TFRicciCD[-a, -b].
** DefTensor: Defining Kretschmann scalar KretschmannCD[].
** DefCovD: Computing RiemannToWeylRules for dim 4
** DefCovD: Computing RicciToTFRicci for dim 4
** DefCovD: Computing RicciToEinsteinRules for dim 4
** DefTensor: Defining weight +2 density Detg[].      Determinant.
```

In[ ]:= **exp = -1 / 4 * epsilon[g][-a, -b, -c, -d] * epsilon[g][e, f, d, m] * RiemannCD[b, c, -e, -f]**

Out[ ]=

$$-\frac{1}{4}\left(-\delta_a{}^m\ \delta_b{}^f\ \delta_c{}^e+\delta_a{}^f\ \delta_b{}^m\ \delta_c{}^e+\delta_a{}^m\ \delta_b{}^e\ \delta_c{}^f-\delta_a{}^e\ \delta_b{}^m\ \delta_c{}^f-\delta_a{}^f\ \delta_b{}^e\ \delta_c{}^m+\delta_a{}^e\ \delta_b{}^f\ \delta_c{}^m\right) R[\nabla]^{bc}{}_{ef}$$

In[ ]:= **ToCanonical[%]**

Out[ ]=

$$g^{mb}\ R[\nabla]_{ab}-\frac{1}{2}\ \delta_a{}^m\ R[\nabla]$$

In[ ]:= **% // RicciToEinstein**

Out[ ]=

$$-\frac{1}{2}\ \delta_a{}^m\ R[\nabla]+g^{mb}\left(G[\nabla]_{ab}+\frac{1}{2}\ g_{ba}\ R[\nabla]\right)$$

In[ ]:= **ToCanonical@ContractMetric[%]**

Out[ ]=

$$G[\nabla]_a{}^m$$